# C++ Notes: Functions

(Creating and Calling Functions)

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## Learning Objectives

After this section, you will be able to:

- Understand what a function is
- Create and call functions
- Pass parameters to functions
- Reuse code using functions
- Use function prototypes (stubs)

## What is a Function?

**Core idea**

A function is a reusable block of code that performs a specific task.

The `main()` function is the entry point of every C++ program.

```
int main() {
 cout << "Program running";
 return 0;
}
```

Anything inside `main()` runs automatically.

## Creating a Simple Function

```
void sayHi() {
 cout << "Hello user" << endl;
}
```

- `void` → return type (returns nothing)
- `sayHi` → function name
- `{}` → function body

## Calling a Function

```
int main() {
  sayHi();
  return 0;
}
```

**Output:**

```
Hello user
```

**Execution Flow**

- Program starts in `main()`
- `sayHi()` is called
- Code jumps to `sayHi()`
- Executes its code
- Returns back to `main()`

## Adding Parameters

```
void sayHi(string name) {
  cout << "Hello " << name << endl;
}
```

Calling the function:

```
int main() {
  sayHi("Mike");
  return 0;
}
```

**Output:**

```
Hello Mike
```

## Multiple Parameters

```
void sayHi(string name, int age) {
  cout << "Hello " << name
      << ", you are " << age
      << " years old." << endl;
}
```

Calling multiple times:

```
int main() {
  sayHi("Mike", 60);
  sayHi("Tom", 45);
  sayHi("Steve", 19);
  return 0;
}
```

**Output:**

```
Hello Mike, you are 60 years old.
Hello Tom, you are 45 years old.
Hello Steve, you are 19 years old.
```

**Why Functions are Powerful**

Write code once → reuse it many times.

## Function Prototype (Stub)

If a function is written below main(), C++ must be told about it first.

```
void sayHi(string name, int age); // prototype

int main() {
  sayHi("Mike", 60);
  return 0;
}

void sayHi(string name, int age) {
  cout << "Hello " << name
      << ", you are " << age
      << endl;
}
```

**Function Prototype**

A prototype declares a function before it is defined.

## Common Mistakes

**Common mistakes**

- Forgetting parentheses when calling a function
- Mismatching parameter types
- Defining function below main() without a prototype
- Forgetting return type

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